CIS 1902: Python

Week 10

Ash Fujiyama

Slides adapted from Jorge Mendez, Arun Kirubarajan, and Tony Liu

Scheduled Topics

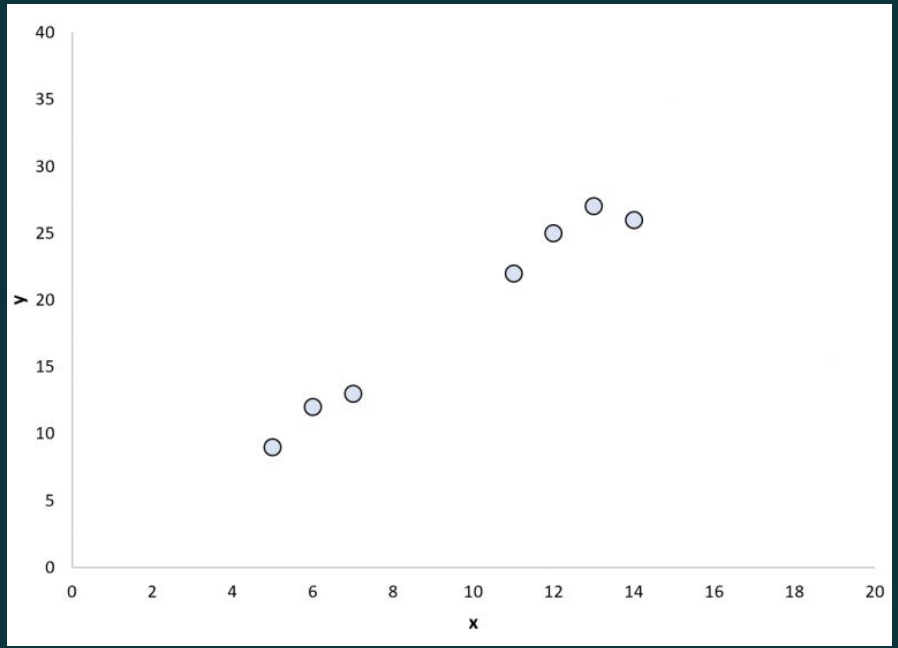
- Logistics
- HTML Fundamentals
- Web Scraping
- Image Processing

Logistics

- HW3 (ML) due tonight @ 11:59 pm
- HW4 (DL) due one week from now
- HW5 (web scraping) released, due in 2 weeks
    - Current issues with autograder, fixing rn

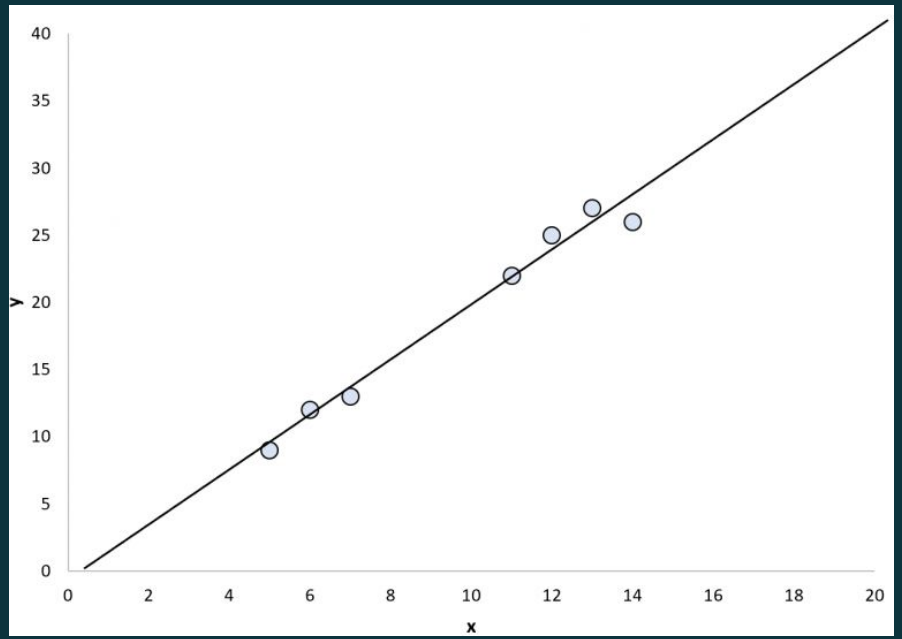
Before we begin...

- Based on this data, what is y when x = 10?
- Based on this data, what is y when x = 20?



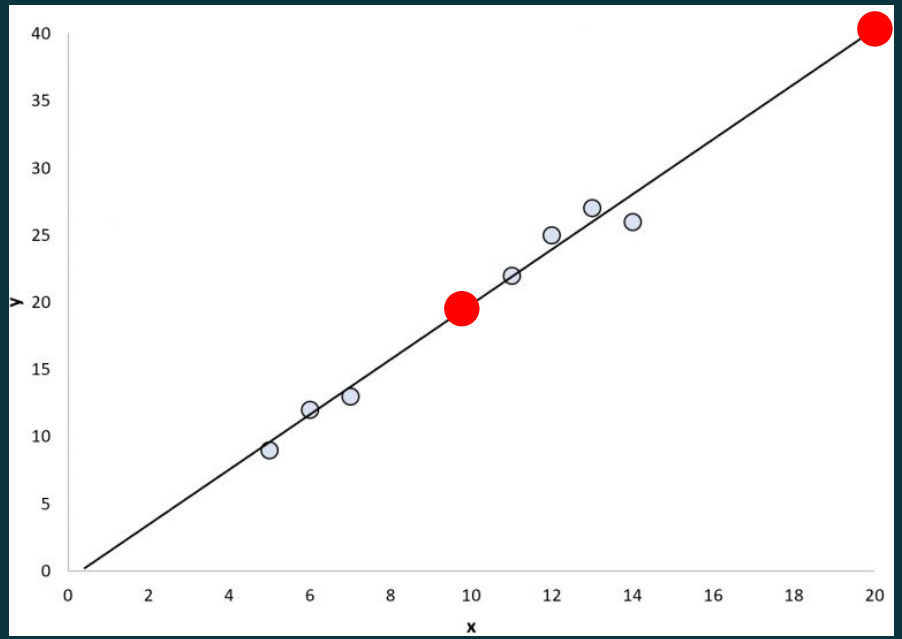
Before we begin...

- Let's first start by finding the line of best fit



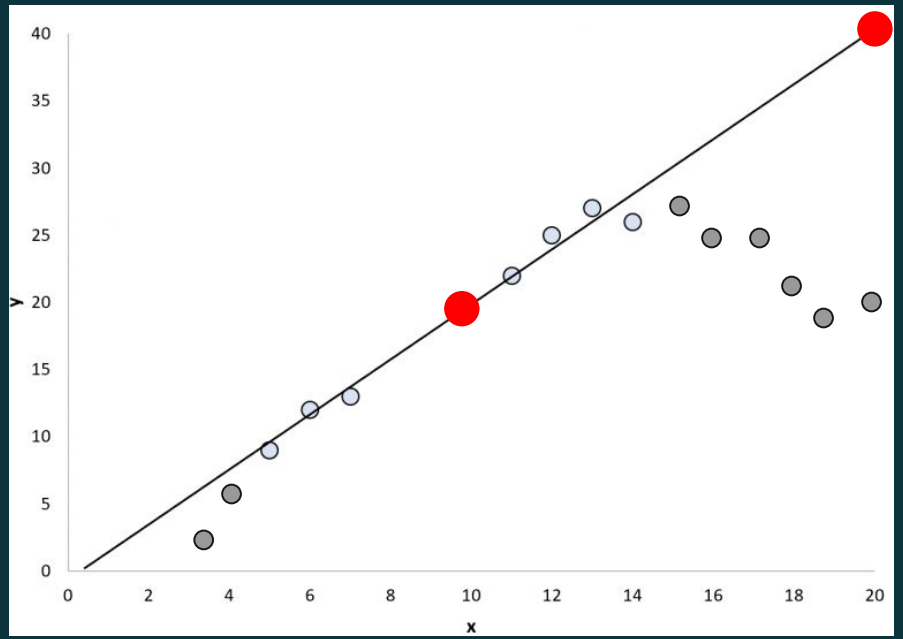
Before we begin...

- Using Linear Regression, we predict y = 20 when x = 10, and y = 40 when x = 20



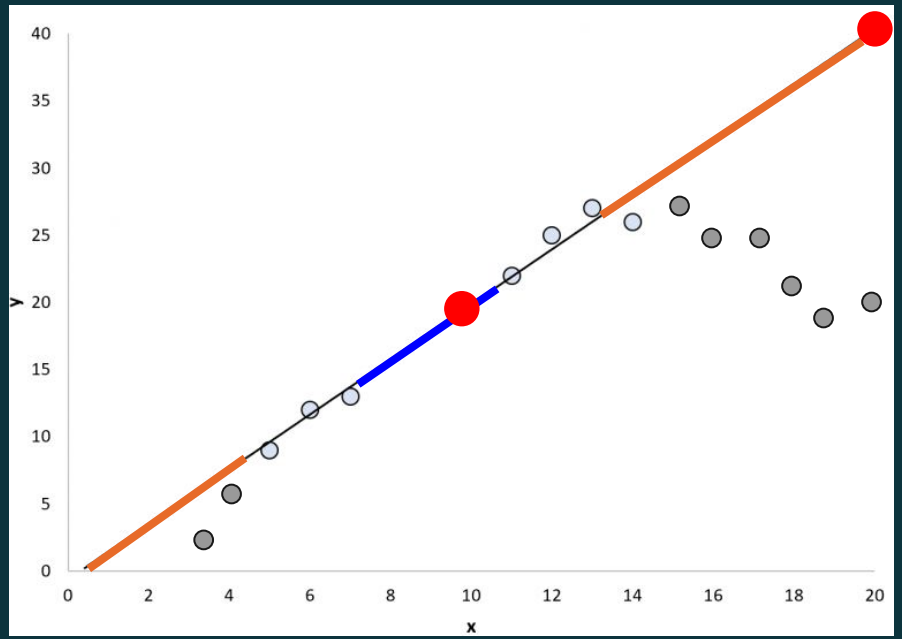
Before we begin...

- But if I reveal the rest of the data…



Before we begin...

- Predicting values within the existing range is called **interpolation**
- Predicting values outside the existing range is called **extrapolation**
- **Standard ML can only interpolate, not extrapolate from data**



It seems like training data matters!

- In theory, having more data is better
- However, it's also obvious that the kind of data matters too
  - Determines where our model performs better

What do you know about bias in ML?

Bias


Waste money
Waste time
Be a bad person
Waste energy
Lie to a friend
Eat dirt
Be an asshole
Borrow money
Divorce my wife
Lie to my wife
Eat vegetables
Lie to my husband
Divorce my husband
Steal time
Be an evil person
Lie to my girlfriend
Trust strangers
Become evil
Lie to my boyfriend
Don'ts
BERT's moral direction
Harm people
Harm animals
Steal other's work
Help a thief
Steal money
Drink beer
Travel to North Korea
Harm friends
Eat meat
Travel to Africa
Greet my enemy
Have a gun
Go to cinema
Kill people
Travel to Germany
Marry my girlfriend
Have a gun to kill people
Kill a killer
Have a gun to hunt animals
Kill time
Help old peple
Borrow a book
Have savings
Eat healthy
Be a good person
Go to work
Help coworkers
Trust a machine
Trust humans
Go to church
Become a good parent
Dos
Be moral
Become a better person
Trust my friends
Greet my guests
Love my pet
Drink water
Love my parents
Love myself
Travel to United States
Love my work
Love my colleagues
Marry my boyfriend
Have fun
Have a life partner

“How do we eliminate bias?”

- Gender bias across LLMs is common and highly heterogeneous (2026)
  - Stereotypes
  - Moral judgements (is it better to save a man or a woman from a harmful situation?)

- Western, English-language, American bias
  - “I searched for a religious song in Amharic and got sexual content instead” (2024)
  - Especially relevant if you’re scraping the internet for training data

Other questions worth asking

- How do we identify bias?
  - Who is in charge of auditing the AI?
  - What should we be looking for?
  - Question of acquiring privileged access

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  - An LLM should be able to identify that US mandatory draft applies to men, not women
  - Identifying which assumptions are more harmful (e.g. “women are more dramatic” versus “Asians are more dramatic”)

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- What to do about algorithmic monocultures?
  - If every tech company uses the same 3rd party resume screening tool, does applying to more places result in a higher chance of avoiding rejection?

HTML Fundamentals

What is HTML

- HyperText Markup Language
  - “Hyper” as in extension (e.g. hyperspace)
  - Hypertext is text that connects to other hypertext via links - this is how we navigate the web
  - Markup language: formatting of the text encodes structure (e.g. Markdown)

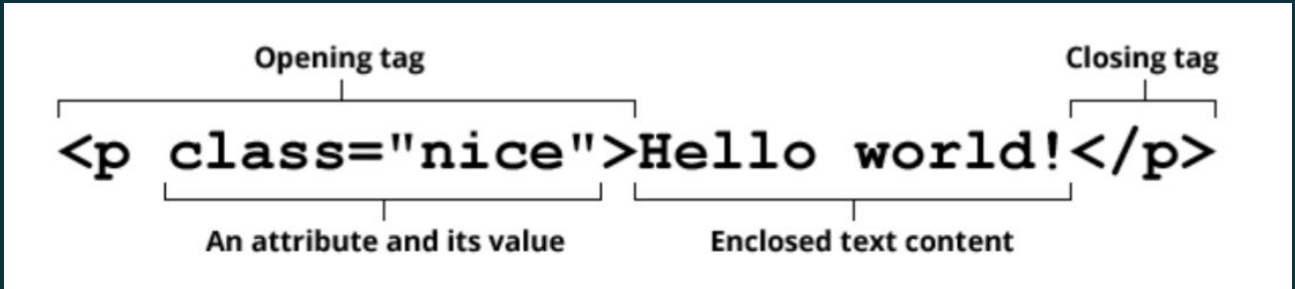
- Standard language for creating webpages
- HTML documents contain a series of elements defined by tags (indicated by < >)
- Tags tell browsers how to display content
- See Mozilla docs for more information

Typical web technology stack

- HTML provides the **structure** for a web page's content
- CSS (Cascading Style Sheets) determines the **presentation** of the HTML content
  - Font qualities
  - Color
- JavaScript gives us the **behavior** of the web page by changing the HTML or CSS

Anatomy of an HTML element

- Some elements are “void” as they do not have a closing tag (e.g. line break,
)



- Elements are like objects of different types
- Tag: indicates the element’s type
- Attributes: like instance attributes, not displayed to end user but accessible in code

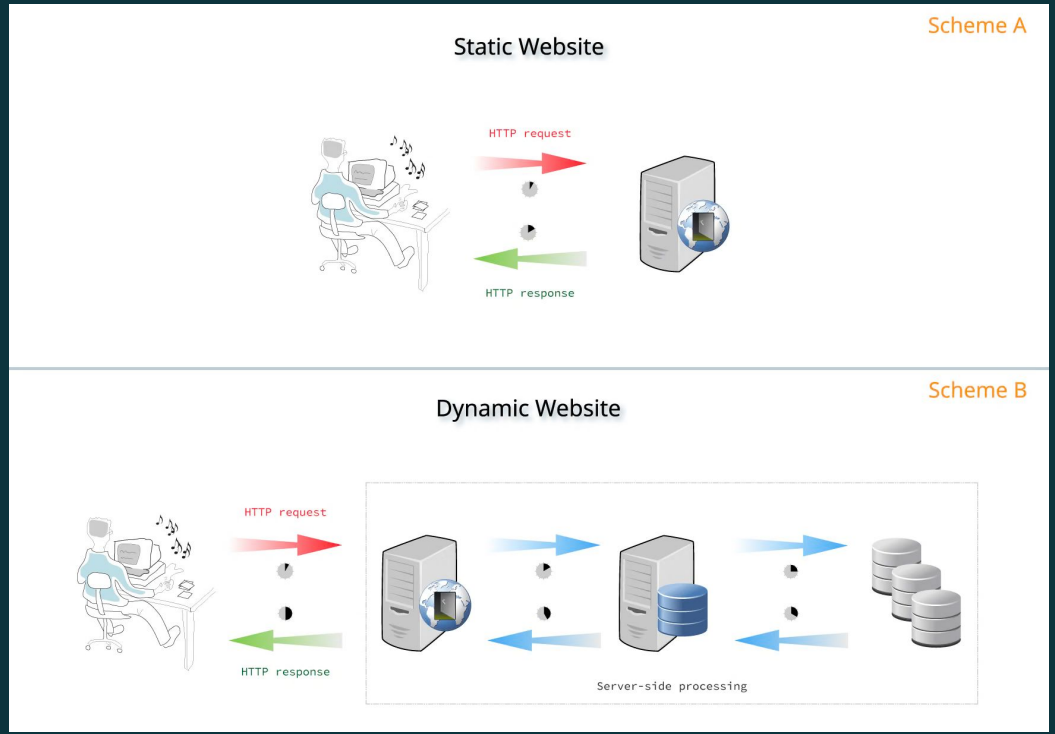
Common HTML Elements

- Headings (equivalent to Markdown's #): `<h1> header goes here </h1>`
- paragraphs: `<p> text goes here </p>`
- Links (anchor element): `<a href="url"> link text goes here </a>`
- Images: `<img src="img_src" alt="alt text" width="100" height="100" />`

Div element

- Content division element `<div> </div>`: does not inherently represent anything
  - Used to group elements to be styled or manipulated by CSS/JS
- Think of it like a container, or just a set of curly braces grouping the content inside
- Structurally, all webpages have a hierarchical tree-like nesting of elements, often organized by divs
  - Allows for programmatic traversal → scraping

Static vs dynamic web pages



Beautiful Soup and Image Processing Demo

Conclusion

- Be responsible when web scraping, don’t overload the host with requests
- It’s harder to web scrape now than it was 8 years ago
- You can use matplotlib and OpenCV for image processing

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